

Project : Sales Data Analysis

1. Retrieve all records from the sales dataset.

```
select * from product_sale;
```

2. Display all unique product categories available in the dataset.

```
select distinct Product_Category from product_sale;
```

3. Find all sales made by a specific Sales Representative (e.g., Alice).

```
select * from product_sale where Sales_Rep ='Alice';
```

4. Get all transactions from the “North” region where the Sales_Amount is above 5000.

```
select * from product_sale where Region = 'North' and Sales_Amount > 5000;
```

5. List all customers who paid via “Credit Card” and are ‘Returning’ customers.

```
select * from product_sale where Payment_Method = 'Credit Card' and  
Customer_Type='Returning';
```

6. Retrieve sales records between ‘2023-01-01’ and ‘2023-06-30’.

```
select * from product_sale where str_to_date(Sale_Date,'%d-%m-%Y') between '2023-01-01'  
and '2023-06-30';
```

7. Calculate the total Sales_Amount for each Region.

```
select Region , round(sum(Sales_Amount)) from product_sale group by Region;
```

8. Find the average Unit_Price and Unit_Cost by Product_Category.

```
select Product_Category, round(avg(Unit_Price),2) , round(avg(Unit_Cost),2) from  
product_sale group by Product_Category;
```

9. Determine the total number of units sold (Quantity_Sold) by each Sales_Rep.

```
select Sales_Rep, sum(Quantity_Sold) as Total_Units from product_sale group by Sales_Rep;
```

10. Identify the highest and lowest Sales_Amount in the dataset.

```
with abc as
```

```
(
```

```
select * from product_sale where Sales_Amount = (select max(Sales_Amount) from  
product_sale)
```

```
union all
```

```
select * from product_sale where Sales_Amount = (select min(Sales_Amount) from  
product_sale)
```

```
)
```

```
select * from abc;
```

11. Find the total discount given across all transactions.
 select Payment_Method, sum(discount) from product_Sale group by Payment_Method;
12. Calculate the average Sales_Amount per month.
 select year(str_to_date(Sale_Date , '%d-%m-%Y')) as Year,
 month(str_to_date(Sale_Date , '%d-%m-%Y')) as month , avg(Sales_Amount) from
 product_Sale group by year(str_to_date(Sale_Date , '%d-%m-%Y')) ,
 month(str_to_date(Sale_Date , '%d-%m-%Y')) order by Year , month;
13. Show the total sales and total quantity sold per Product_Category and Region.
 select Product_Category , round(sum(Sales_Amount),2) as total_sale, sum(Quantity_Sold) as
 total_qty from product_Sale group by Product_Category;
14. Calculate profit for each transaction
 select * , round((Unit_Price-Unit_Cost)*Quantity_Sold,2) as profit from product_Sale order
 by profit desc limit 10;
15. Find the best-performing Sales Representative based on total Sales_Amount.
 select Sales_Rep, sum(Sales_Amount) as Total_Sales from product_Sale group by Sales_Rep
 order by Total_Sales desc limit 1;
16. Find the region with the highest total profit.
 select Region, round(sum((Unit_Price-Unit_Cost)*Quantity_Sold),2) as profit from
 product_Sale group by Region order by profit desc limit 1;
17. Get the top 5 products (Product_ID) with the highest total sales amount.
 select Product_ID,Product_Category, round(sum(Sales_Amount),2) as Total_sales from
 product_sale group by Product_ID, Product_Category
 order by Total_sales desc limit 5;
18. Find the average discount offered per Product_Category.
 select distinct Product_Category, round(avg(Discount) over(partition by
 Product_Category),2) as avg_discount from product_sale;
 select Product_Category , round(avg(Discount),2) as avg_discount from product_sale group
 by Product_Category;
19. Identify which Sales_Channel (Online vs Retail) generates more revenue.
 select Sales_Channel , round(sum(Sales_Amount),2) as total_sale from product_sale group
 by Sales_Channel order by total_sale desc;
20. For each Sales_Rep, calculate their average Sales_Amount per transaction.
 select Sales_Rep, Payment_Method , round(avg(Sales_Amount),2) as Avg_sale from
 product_sale group by Sales_Rep, Payment_Method order by Payment_Method;
21. Rank Sales Representatives by total revenue using a window function.
 select Sales_Rep, sum(Sales_Amount) as Total_Sales , rank() over(order by
 sum(Sales_Amount) desc) as Revenue_Rank from product_sale
 group by Sales_Rep;

22. Find the percentage contribution of each Region to total sales.
 select Region , sum(Sales_Amount) as Total_Sale ,
 round((sum(Sales_Amount) / (Select sum(Sales_Amount) from product_sale))*100,2) as
 percent_contribution from product_sale group by Region order by percent_contribution
 desc;

23. Display monthly sales trends (Month vs Total Sales).
 select date_format(str_to_date(Sale_Date, '%d-%m-%Y'), '%M') as Month_Name ,
 sum(Sales_Amount) as Total_Sales from product_sale
 group by Month_Name order by Month_Name;

24. Identify any Region-Sales_Rep combinations where the total discount given exceeds 20% of
 total sales.
 select Region , Sales_Rep , sum(Sales_Amount) as Total_sale , sum(Sales_Amount *
 discount) as total_discount, round(sum(Sales_Amount * discount) / sum(Sales_Amount) *
 100 ,2) as Discount_percentage from product_sale group by Region , Sales_Rep
 having sum(Sales_Amount * discount) > 0.2 * sum(Sales_Amount);

25. Compare the average sales amount between New and Returning customers.
 select Customer_Type , round(avg(Sales_Amount),2) from product_sale where
 Customer_Type in ('Returning' , 'new') group by Customer_Type;

26. Find the most frequently sold Product_Category for each Region.
 select region , Product_Category from
 (select Region , Product_Category , count(*) as total_sale, rank() over(partition by Region
 order by count(*) desc)
 as rnk from product_sale group by Region, Product_Category)as ranked where rnk = 1;

27. Calculate profit margin (%) for each Product_Category (Total Profit / Total Sales_Amount) *
 100.
 select Product_Category , round(sum((Unit_Price-Unit_Cost)*Quantity_Sold),2) as
 Total_Profit , round(sum(Sales_Amount),2) as Total_amount ,
 round((sum((Unit_Price-Unit_Cost)*Quantity_Sold) / sum(Sales_Amount)) * 100 , 2) as
 Profit_margin
 from product_sale group by Product_Category;

28. Find the top-performing month for each Sales Representative.
 select * from (select Sales_Rep , date_format(str_to_date(Sale_Date , '%d-%m-%Y'), '%M
 %Y') as month_Name, sum(Sales_Amount) as total_Sale, rank() over(partition by Sales_Rep
 order by sum(Sales_Amount) desc) as rnk from product_sale group by Sales_Rep ,
 date_format(str_to_date(Sale_Date , '%d-%m-%Y'), '%M %Y')) as ranked where rnk=1 order
 by Sales_Rep desc ;

29. Create a summary table showing each Region, total sales, total profit, and average discount.
 select Region , round(sum(Sales_Amount),2) as Total_sale , round(sum((Unit_Price-
 Unit_Cost)*Quantity_Sold),2) as Total_profit , round(avg(Discount),2) as Average_discount
 from product_sale group by Region;